

AROMAL DILEEP

Munnar, Kerala, India

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[LinkedIn](#) | [GitHub](#) | [Kaggle](#) | [Leetcode](#)

PROFESSIONAL SUMMARY

Self-driven Machine Learning Engineer specializing in building production-ready AI systems. Strong hands-on experience in Computer Vision, Semantic Search, and Deep Learning, with a proven ability to design scalable backend architectures and deploy end-to-end ML pipelines. Focused on solving real-world problems through efficient and practical AI solutions.

TECHNICAL SKILLS

Programming:

Python, SQL

Machine Learning & AI:

Deep Learning (CNNs, GANs), NLP, Computer Vision, Semantic Search, Generative AI

Frameworks & Libraries:

PyTorch, TensorFlow, Scikit-learn, Hugging Face, OpenCV

Backend & Deployment:

FastAPI, Flask, Streamlit

Tools & Infrastructure:

Linux, Git, Docker, AWS, Azure, FAISS, PostgreSQL, Optuna

KEY PROJECTS

SmartCart AI – Multimodal Semantic Search

[GitHub](#)

Python, PyTorch, CLIP, FAISS, FastAPI, Docker

- Built a production-ready multimodal search system using CLIP embeddings, enabling unified text and image-based product retrieval.
- Implemented FAISS-based vector similarity search across ~60K products, achieving fast and scalable retrieval performance.
- Designed and deployed RESTful APIs for seamless integration into real-world e-commerce systems.
- Developed evaluation pipelines using LLM-assisted relevance scoring and ranking metrics (Precision@K) to measure search quality.

Face Recognition Based Attendance System

[GitHub](#)

Python, OpenCV, Face Recognition

- Developed a real-time face detection and recognition pipeline for automated attendance tracking.
- Integrated camera hardware with an optimized inference pipeline to ensure low-latency processing.
- Built a centralized database system with automated reporting and export capabilities.

Vanilla GAN & Generative Research

PyTorch, Optuna, Diffusion Models

- Implemented a Generative Adversarial Network from scratch with hyperparameter optimization using Optuna.
- Conducted comparative analysis between GANs and Latent Diffusion models to evaluate architectural trade-offs.
- Explored text-to-image generation pipelines and underlying model design strategies.

ACHIEVEMENTS & COMPETENCIES

- **Kaggle Expert (Top ~3.5%)** with multiple medals and high-quality notebook contributions
- Built AI-powered applications and scalable backend systems through freelance projects
- Strong foundation in Data Structures, Algorithms (DSA), and Object-Oriented Programming
- Experience deploying end-to-end ML systems using cloud platforms and containerization

EDUCATION & LEARNING

Independent Machine Learning Engineer (Self-taught)

- Completed structured learning in Machine Learning, Deep Learning, Computer Vision, NLP, and Generative AI
- Hands-on development of production-grade AI systems and research-oriented projects